



Faculty of Engineering and Technology
Electrical and Computer Engineering Department

ARTIFICIAL INTELLIGENCE

ENCS3340

Project Report

**Comparative Study of Image Classification Using Decision
Tree, Naive Bayes, and Feedforward Neural Networks**

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Introduction

Our project focuses on implementing and comparing different machine learning models for image classification. The goal is to evaluate the performance of Naive Bayes, Decision Tree, and Feedforward Neural Network (MLP) classifiers on a labeled image dataset. Images were preprocessed by resizing, and flattening into 1D vectors to serve as input features. The models were trained and tested using standard evaluation metrics such as accuracy, precision, recall, F1-score, and confusion matrix. By analyzing the results, the project highlights the strengths and weaknesses of each approach, helping us understand which model works best for image classification.

Dataset description

In this project, we used a subset of the Intel Image Classification dataset, which was originally sourced from Kaggle. The full dataset contains approximately 14,034 images categorized into six natural scene classes: buildings, forest, glacier, mountain, sea, and street. For our analysis, we focused on three of these classes: street, glacier, and forest.

Each image in the dataset describe a natural or urban scene and is stored in a folder corresponding to its label. The original image sizes vary, so to ensure consistency and simplify the training process, we resized all images to 32×32 pixels and converted them to grayscale and normalized it.

Class name	street	glacier	forest
Number of images	2382	2404	2271

Explanations of models

1. Naive Bayes Classifier

Naive Bayes is a fast and simple algorithm that uses probability to decide which class an image belongs to. It treats each pixel as if it's unrelated to the others (that's why it's called "naive"). In the code: We used GaussianNB() to create and train the model:

```
#train naive bayes model
nb_model = GaussianNB().fit(X_train, y_train)
```

2. Decision Tree Classifier

Decision Trees make decisions like a flowchart. It checks pixel values and asks “yes/no” questions (for example: is pixel 120 > 0.5?). Based on the answers, it moves down the tree until it decides which class the image belongs to. In the code: We used DecisionTreeClassifier() to train the model:

```
#train decision tree model
dt_model = DecisionTreeClassifier().fit(X_train, y_train)
```

3. Feedforward Neural Network (MLP - Multi-Layer Perceptron)

This is a neural network that has multiple layers (128 and 64 neurons in our case). It learns more complex patterns in the data. Each neuron looks at part of the image and learns what features are important for classification. In the code: We used MLPClassifier() with hidden layers:

```
#train MLP neural net model with two hidden layers: 128 and 64 neurons
mlp_model = MLPClassifier(hidden_layer_sizes=(128, 64), max_iter=1000, random_state=42).fit(X_train, y_train)
```

Evaluation Results and Discussion

1. Accuracy Comparison

```
Accuracy Comparison
=====
Naive Bayes      : 0.6827
Decision Tree    : 0.5949
MLP Neural Net   : 0.7373
```

MLP Neural Network achieved the highest accuracy overall. Naive Bayes performed moderately, while the Decision Tree had the lowest accuracy.

2. Precision, Recall, and F1-Score Per Class

A- Naïve Bayes

```
Naive Bayes
=====
Class: forest
Precision: 0.70
Recall    : 0.80
F1-score  : 0.74

Class: glacier
Precision: 0.70
Recall    : 0.76
F1-score  : 0.73

Class: street
Precision: 0.64
Recall    : 0.49
F1-score  : 0.55
```

Naive Bayes worked better for "forest" and "glacier" but struggled with "street". Lower recall for "street" shows it often missed those images.

B- Decision Tree

```
Decision Tree
=====
Class: forest
Precision: 0.64
Recall   : 0.63
F1-score : 0.64

Class: glacier
Precision: 0.63
Recall   : 0.65
F1-score : 0.64

Class: street
Precision: 0.51
Recall   : 0.50
F1-score : 0.51
```

The Decision Tree showed balanced but low scores for all classes. It had trouble separating the classes clearly, especially "street".

C- MLP Neural Net

```
MLP Neural Net
=====
Class: forest
Precision: 0.73
Recall   : 0.76
F1-score : 0.74

Class: glacier
Precision: 0.77
Recall   : 0.83
F1-score : 0.80

Class: street
Precision: 0.70
Recall   : 0.63
F1-score : 0.66
```

MLP gave the best results, especially for "glacier" and "forest". It was also more balanced and reliable across all three classes.

3. Confusion Matrice

Note : Each row represents the actual class, and each column represents the predicted class.

A- Naïve Bayes

Naive Bayes Confusion Matrix			
	forest	glacier	street
forest	381	42	53
glacier	42	357	72
street	125	114	226

Naive Bayes often confused "street" with "forest" and "glacier". It performed well for "forest", but had many misclassifications.

B- Decision Tree

Decision Tree Confusion Matrix			
	forest	glacier	street
forest	301	70	105
glacier	48	307	116
street	120	113	232

Decision Tree confused all three classes frequently. Misclassifications were high, especially for "street" and "forest".

C- MLP Neural Net

MLP Neural Net Confusion Matrix			
	forest	glacier	street
forest	360	39	77
glacier	33	389	49
street	98	75	292

MLP had fewer misclassifications and stronger separation between classes. "Glacier" had the best recognition, with very few wrong predictions.

❖ The Over All results

Model	Accuracy	Strengths	Weaknesses
Naive Bayes	68.27%	Fast, simple, good starting point	Assumes independence between pixels
Decision Tree	59.49%	Easy to understand and interpret	Overfits, low generalization
MLP Neural Net	73.73%	Best accuracy, learns complex patterns	Needs more time/resources to train

For this image classification project, the **MLP Neural Network is the most effective model**, since it successfully captures complex visual features, reduces confusion among classes, and achieves the highest accuracy. Although it's more resource-intensive, the performance benefits make it the best choice in this case.